Table 2. Complete results on the Soccer-GMR test set, with MLLM references

Our 14 runs: 1,036 queries (544 positive / 492 negative), τ = 0.5. Six paper Table 4 rows retain τ = 0.4. All values: %.

Model	*Null-Set Rejection*		*Temporal Localization*				*Overall GMR*	
	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3
Moment-DETR	71.51	1.21	7.69	4.48	10.87	1.14	6.45	3.05
QD-DETR	74.84	19.71	16.26	10.80	22.01	11.55	17.52	11.56
CG-DETR	74.83	6.65	18.32	12.09	25.59	9.52	14.13	8.12
EaTR	**77.85**	24.48	16.82	10.83	23.42	11.57	18.92	12.99
Moment-DETR-GMR	70.20	67.94	6.68	4.64	9.44	0.88	42.50	39.39
QD-DETR-GMR	<u>77.01</u>	71.48	14.62	9.47	20.17	7.43	46.49	41.47
CG-DETR-GMR	76.40	57.47	18.65	12.65	24.90	11.98	34.35	29.14
EaTR-GMR	76.56	65.60	16.20	10.64	21.38	9.59	40.85	36.00
FlashVTG-GMR	74.30	66.35	<u>25.43</u>	<u>15.17</u>	<u>34.94</u>	<u>17.86</u>	43.57	38.27
Moment-DETR-GMR + S4_minus_Fusion	70.26 (+0.06)	75.00 (+7.06)	7.59 (+0.91)	3.46 (-1.18)	12.40 (+2.96)	0.98 (+0.10)	51.92 (+9.42)	**48.97** (+9.58)
QD-DETR-GMR + S4_minus_Fusion	76.21 (-0.80)	74.21 (+2.73)	16.18 (+1.56)	9.75 (+0.28)	23.77 (+3.60)	12.26 (+4.83)	51.77 (+5.28)	47.02 (+5.55)
CG-DETR-GMR + S4_minus_Fusion	76.72 (+0.32)	<u>75.30</u> (+17.83)	19.33 (+0.68)	13.08 (+0.43)	25.77 (+0.87)	11.87 (-0.11)	<u>53.60</u> (+19.25)	<u>48.79</u> (+19.65)
EaTR-GMR + S4_minus_Fusion	76.03 (-0.53)	**75.32** (+9.72)	17.01 (+0.81)	11.44 (+0.80)	22.48 (+1.10)	9.63 (+0.04)	53.12 (+12.27)	48.53 (+12.53)
FlashVTG-GMR + S4_minus_Fusion	75.15 (+0.85)	74.80 (+8.45)	**25.84** (+0.41)	**15.98** (+0.81)	**35.86** (+0.92)	**18.88** (+1.02)	**53.94** (+10.37)	<u>48.79</u> (+10.52)
MLLM references: paper Table 4 (reported at τ = 0.4)								
Qwen3-VL-4B [Table 4]	47.66	11.54	2.15	1.65		0.12		
Qwen3-VL-8B [Table 4]	52.60	43.11	1.94	1.54		0.16		
Qwen3-VL-32B [Table 4]	57.75	54.26	2.76	2.42		0.06		
TRACE [Table 4]	50.00	0.00	1.40	1.39		0.00		
TimeChat [Table 4]	50.85	4.72	0.49	0.39		0.00		
Qwen3-VL-4B (GRPO) [Table 4]	53.21	13.83	2.91	1.97		1.18		

First 14 rows: our baseline / GMR / GMR + S4_minus_Fusion runs. Bold: best; underline: second-best among reported values.

All S4_minus_Fusion rows use epoch 4. FlashVTG-GMR uses the archived E0 run; its S4 variant uses the historical no-QCD epoch 4.

Our Rej-F1 and G-mIoU use τ = 0.5; AUROC is threshold-free; localization uses positives. No adapter-free FlashVTG run.

Parentheses: S4_minus_Fusion minus same-backbone GMR, in percentage points (pp); negative values indicate decreases.

Last 6 rows: published Table 4 values at τ = 0.4, not our reruns. Unreported mR@5 and G-mIoU cells are blank.